

Flow Termination: the t^* Formula, the Discovery Monovariant and the Damped Billiard

Theorem monograph T06 of the chess-dynamics-lab program —
the KLEIN layer

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Keywords: discrete flow, termination, t^* , billiard, braking

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Setting

The KLEIN layer is the flow layer of the model: a particle with a fixed integer impulse (a, b) moves over the $W \times H$ board grid by the steps $p \mapsto p + (a, b)$. It is the direct heir of Certificate E of the parent monograph~“The Dynamic Principle”, where the termination formula $t^* = \text{lcm}(W/\text{gcd}(a, W), H/\text{gcd}(b, H))$ was proved and reproduced in five languages. Here we carry the theorem over to the chess board, supplement it with the discovery monovariant, and prove a new layer — the billiard with reflections and the braking $\gamma = \delta^4/k$, where $\delta = \pi/n$ is the calibration of the parent monograph (the torus: $n = 4, k = 1, \gamma = \pi^4/256 \approx 0.3805$).

Theorem

Theorem T06 (flow termination). (i) For a particle with impulse (a, b) on the torus $W \times H$ the first-return time is $t^* = \text{lcm}(W/\text{gcd}(a, W), H/\text{gcd}(b, H))$, and the flow visits exactly t^* distinct squares. (ii) The discovery monovariant (the cardinality of the set of visited squares) grows strictly up to t^* and stops. (iii) For the billiard with mirror reflections and the braking $\gamma \in (0, 1)$ multiplying each velocity component by γ at every step, the total path over n steps equals $|v_0|(1 - \gamma^n)/(1 - \gamma)$ and converges to $|v_0|/(1 - \gamma)$: the particle stops within a finite total path.

Proof

(i) The particle position is a pair of coordinates modulo W and H respectively. A return requires simultaneously $t \cdot a \equiv 0 \pmod{W}$ and $t \cdot b \equiv 0 \pmod{H}$. The equation $ta \equiv 0 \pmod{W}$ has solutions that are multiples of $W/\text{gcd}(a, W)$: indeed, the

least positive t with $W \mid ta$ is $W/\gcd(a, W)$, because ta is divisible by W iff t is divisible by $W/\gcd(a, W)$. Likewise for the second coordinate. The common solution is the multiples of the least common multiple of the two periods, that is $t^* = \text{lcm}(W/\gcd(a, W), H/\gcd(b, H))$. The trajectory up to the return is a clean orbit without repetitions: if a square repeated earlier than t^* , the difference of the two moments would give a smaller return period, contradicting minimality. Hence exactly t^* squares are visited.

(ii) The discovery monovariant $D(t)$ — the cardinality of the set of squares visited by the moment t . By (i) the first t^* positions are pairwise distinct, so D grows by one at every step: $D(t) = t$ for $t \leq t^*$ and $D(t) = t^*$ afterwards — a strictly increasing bounded monovariant reaching its maximum exactly at the termination moment. This is the chessboard analogue of the discovery monovariant of Certificate E of the parent monograph.

(iii) The billiard. A reflection at a vertical wall flips the sign of the horizontal velocity component and keeps its magnitude; equivalently, the reflecting trajectory is a straight trajectory on the unfolded torus $2W \times 2H$. The braking multiplies both velocity components by γ after every step, so the speed at step n is $|v_0|\gamma^{n-1}$ and the step covers the path $|v_0|\gamma^{n-1}$ (reflections do not change the step length). The total path over n steps is the geometric sum $|v_0|(1 + \gamma + \dots + \gamma^{n-1}) = |v_0|(1 - \gamma^n)/(1 - \gamma)$, which converges to $|v_0|/(1 - \gamma)$ as $n \rightarrow \infty$, because $0 < \gamma < 1$. Numerically, at the torus calibration $\gamma = \pi^4/256$ the protocol confirms the ratio of the accumulated path to the limit to 10^{-9} at the rest position. The theorem is proved.

Protocol data

Flow certificates (check C4)

Case (W, H, a, b)	t^*	Simulation
(8, 8, 1, 1)	8	8 steps, 8 squares
(8, 8, 3, 5)	8	8 steps, 8 squares
(8, 8, 2, 2)	4	4 steps, 4 squares
(8, 8, 1, 2)	8	8 steps, 8 squares
(8, 8, 1, 0)	8	8 steps, 8 squares
(48, 48, 1, 1)	48	48 steps, 48 squares (E reference)
(24, 36, 3, 5)	72	72 steps, 72 squares (E reference)
(12, 12, 4, 6)	6	6 steps, 6 squares (E reference)
(7, 14, 1, 1)	14	14 steps, 14 squares (E reference)
Billiard (8x8, v=(3,2))	-	path/limit = 1.000000000

Summary of what is proved

We proved three flow results: the exact termination formula t^* carried fully over to the chessboard, the strict discovery monovariant stopping exactly at t^* , and the convergence of the damped billiard with the braking $\gamma = \pi^4/256$ to the finite path

$|v_0|/(1 - \gamma)$. The mechanisms are the Chinese remainder theorem in the lcm-period form, monovariant analysis and the geometric series sum. The KLEIN layer has now the exact termination — the direct heritage of Certificate E.

Verification

Check: `python3 dynamics.py -run C4` or the C4 line of the polyglot core (9 cases of t^* with the simulation; the core is floating-point free, the billiard is verified in the laboratory). Every case is checked both by the formula and by a flow run with the visited-square count; PASS in all seven implementations.